A brief overview on simple returns and log returns in financial data

Simon Leung 8 min read · Dec 5, 2024

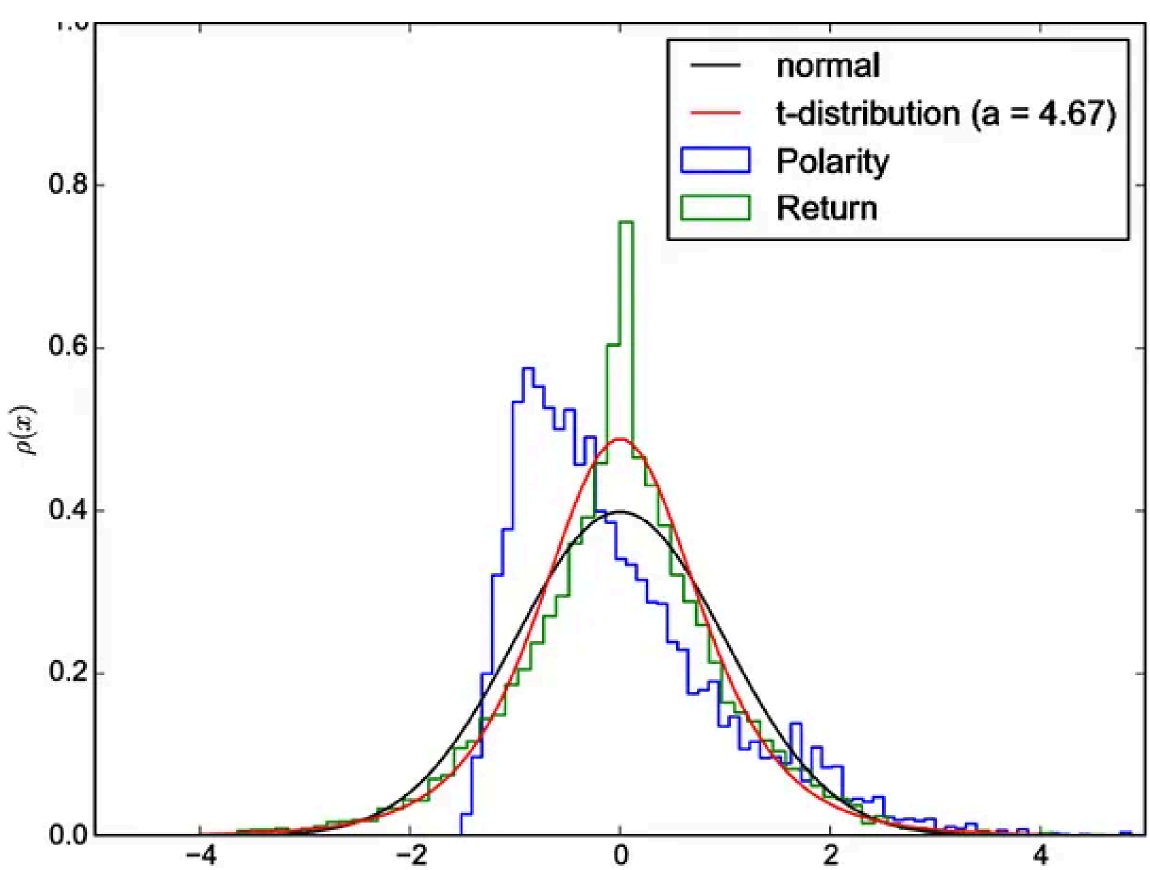


Image from https://www.researchgate.net/figure/Distribution-of-returns-polarities-as-well-as-the-normalized-Gaussian-distribution-and_fig1_318528009

Why It (Returns) Matters

- **Investment Performance**: The return gives investors insight into how well their investment has performed. A positive return indicates a gain, while a negative return indicates a loss.
- **Risk Assessment**: Returns also help in assessing the risk of an investment. Volatile price movements can lead to high returns or significant losses.

Simple Return:
It measure the gain or loss generated on an investment relative to the amount of money invested. It's usually expressed as a percentage. Returns can be calculated over various periods (daily, monthly, yearly).

$$r_i = \frac{P_t}{P_{t-1}} - 1$$

Log Return:
It used to normalize returns and are calculated as the natural logarithm of the ratio of the current price to the previous price. Log returns are particularly useful for statistical analysis because they are time-additive, meaning the log return over multiple periods is simply the sum of the log returns over the individual periods.

$$r_i = Ln(P_t / P_{t-1})$$

Calculation Example 1:
Let's calculate the cumulative returns using both methods for an asset that starts at $100 and experiences returns of 10%, 5%, and -2%.

Simple Returns

1. **Initial Price:** $100
2. **First Period Return:** 10%
   New Price: $100\times(1+0.10)=110$
3. **Second Period Return:** 5%
   New Price: $110\times(1+0.05)=115.5$
4. **Third Period Return:** −2%
   New Price: $115.5\times(1-0.02)=113.19$

Cumulative Simple Return: (113.19–100)/ 100 = 0.1319 or 13.19%

Log Returns

Log returns are additive over multiple periods, so we calculate each period's log return and then sum them up.

1. **First Period Return:** $\ln(110/100)\approx0.0953$
2. **Second Period Return:** $\ln(115.5/110)\approx0.0488$
3. **Third Period Return:** $\ln(113.19/115.5)\approx-0.0201$

Cumulative Log Return: $0.0953+0.0488-0.0201=0.124$
Equivalent Simple Return = e^(0.124) -1 ≈0.1319 or 13.19%

Calculation Example 2:
Let's calculate the cumulative returns for an asset with prices starting at $100, then moving to $110, $90, and finally $120, using both simple and log returns.

Simple Returns

1. **Initial Price**: $100
2. **First Price Movement to $110:**
   Return: (110−100) /100 = 0.10 or 10%
3. **Second Price Movement to $90:**
   Return: (90−110) /110= −0.1818 or -18.18%
4. **Third Price Movement to $120:**
   Return: (120−90)/ 90 = 0.3333 or 33.33%

Cumulative Simple Return:
Cumulative Return=(1+0.10)×(1−0.1818)×(1+0.3333)−1

=1.10 × 0.8182 × 1.3333 − 1

=1.2004 − 1

=0.2004 or 20.04%

Log Returns

Log returns are additive over multiple periods. We calculate the log return for each price change and sum them up.

1. **First Log Return:** ln(110/100)=ln(1.10)≈0.0953

2. **Second Log Return:** ln(90/110)=ln(0.8182)≈−0.2007
3. **Third Log Return:** ln(120/90)=ln(1.3333)≈0.2877

Cumulative Log Return:

Cumulative Log Return=0.0953+(−0.2007)+0.2877=0.1823
Equivalent Simple Return=e^(0.1823)−1≈0.2004 or 20.04%

Both simple and log returns have their advantages and disadvantages.

Simple Returns

Pros:

1. **Simplicity:** Easy to understand and calculate.
2. **Common Usage:** Widely used in reports and by investors for quick assessments.
3. **Relative Changes:** Provides a straightforward percentage change, making it easy to interpret performance over time.
4. ***Asset-additive***: Portfolio return is the weighted average of the stocks in the portfolio.

$$R_{p,t} = \sum_{i=1}^{n} w_{i,t} R_{i.t}$$
,

where weights are the proportion of an individual stock in the portfolio and the sum of the weights is 1.

Cons:

1. **Non-*time*-Additive**: Cannot simply add returns over multiple periods; compounded returns require more complex calculations.
2. **Distortion Over Time**: Over long periods, the multiplicative nature can cause significant distortions in return calculations.
3. **Extreme Values:** More susceptible to being skewed by outliers or extreme values in the data.

Log Returns

Pros:

1. ***Time*-Additivity**: Log returns can be added over multiple periods, which simplifies analysis and modeling.
2. **Normal Distribution:** Often closer to a normal distribution, making them more suitable for statistical methods and models.
3. **Compounding:** Better accounts for the compounding effect over time, providing a more accurate measure for continuous growth rates.
4. **Zero Boundary:** Log returns naturally bound asset prices at zero, which aligns with the real-world behavior of prices not going below zero.

Cons:

1. **Complexity:** More complex to understand and calculate compared to simple returns.
2. **Less Intuitive:** Not as intuitive for non-technical investors, as the results are not in straightforward percentage terms.
3. **Approximation:** Log returns are an approximation and might not be as accurate for small changes in value compared to simple returns.

4. **Not asset-additive.** The weighted average of log returns of individual stocks is not equal to the portfolio return. In fact, log returns are not a linear function of asset weights.

If we want to model returns using the normal distribution!

- SIMPLE RETURNS: The product of normally distributed variables is NOT normally distributed
- LOG RETURNS: The sum of normally distributed variables DOES follow a normal distribution

Does it make sense to assume that prices are log normally distributed?

A log-normal distribution means that the natural logarithm of the price follows a normal distribution. This assumption is rooted in the idea that prices cannot go negative and that returns are compounded multiplicatively rather than additively.

Limitations

- **Fat Tails:** Real-world financial returns often exhibit "fat tails" , skewness, and higher kurtosis, meaning they have more extreme values than the normal distribution predicts. This can lead to underestimating the probability of extreme price movements. Financial analysts often use more sophisticated models, like the t-distribution or non-parametric methods, to capture these characteristics better.

- **Mean Reversion:** In certain financial instruments or over long periods, prices may exhibit mean-reverting behavior, which a log-normal distribution does not capture.

> *Fat Tail Risk*
>
> *Let's start with the very important concept of fat tail risk (also known as kurtosis risk). Fat tail risk is the risk that arises when you assume a normal distribution on an observation that, in reality, has much fatter tails than a normal distribution. In other words, it is the risk that arises from systematically underestimating the likelihood of large deviations from the mean (typically to the downside).*

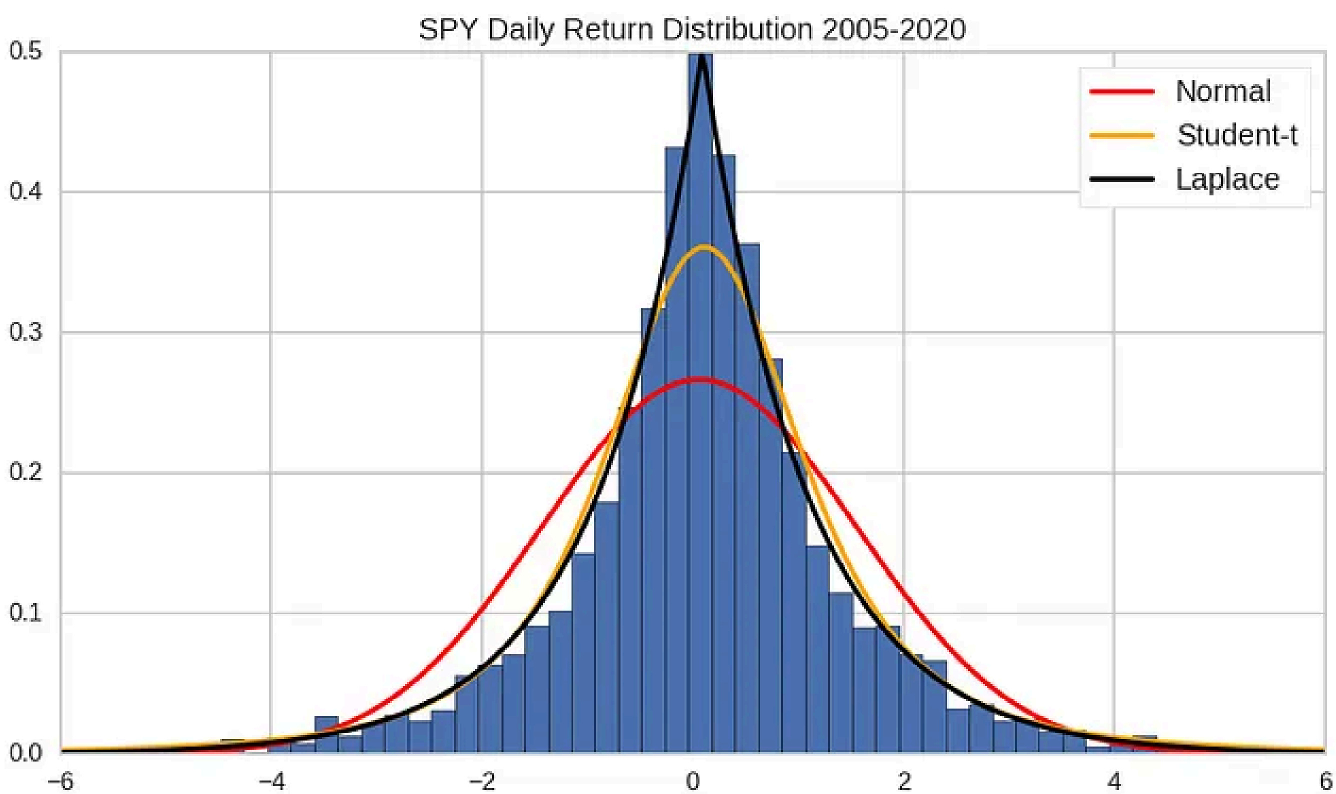


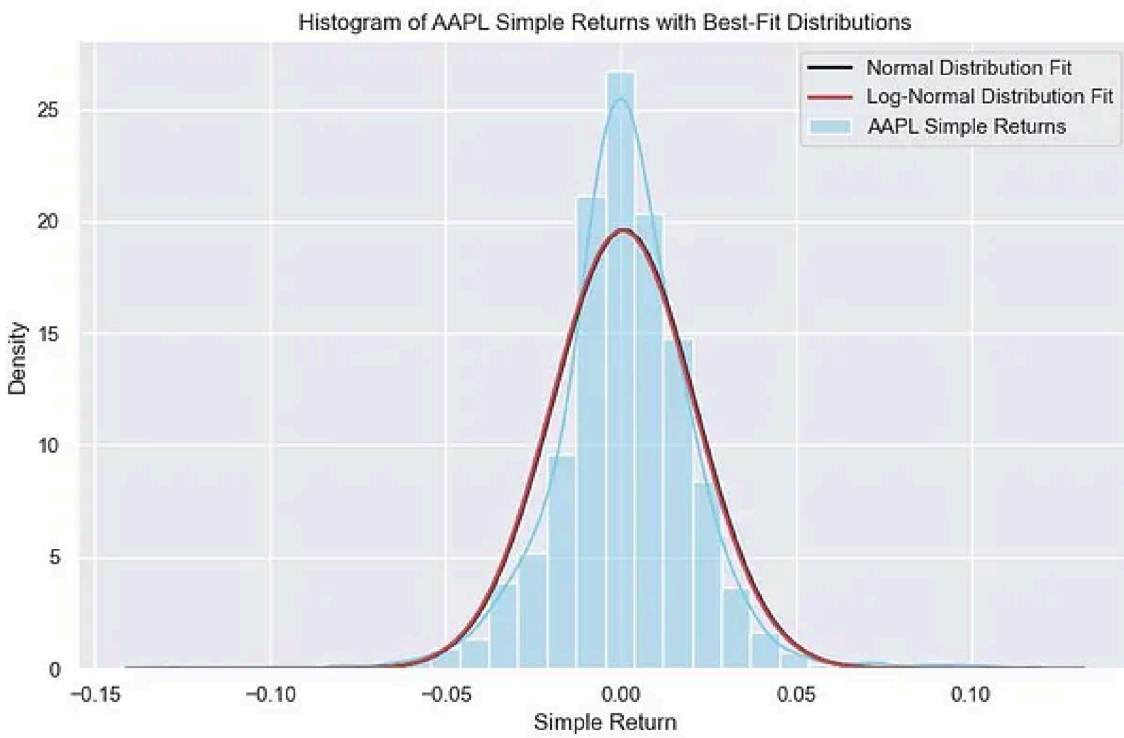
Image and quote from https://tradeoptionswithme.com/probability-dsitribution-of-stock-market-returns/

The returns of AAPL stock's closing prices (2020 to 2024) fit better with a Student's t-distribution (not Cauchy or Laplace distribution)!

```
import yfinance as yf
# Fetch AAPL stock data
AAPL_data = yf.download("AAPL", start="2020-01-01", end="2024-10-01")
# Calculating Simple returns
AAPL_data['Simple Returns'] = AAPL_data['Close'].pct_change()
AAPL_data.dropna(inplace=True)
# Calculating log returns
AAPL_data['Log Returns'] = np.log(AAPL_data['Close'] / AAPL_data['Close'].shift(
# Dropping NaN values resulting from the shift
AAPL_data.dropna(inplace=True)
```

```
import matplotlib.pyplot as plt
import pandas as pd
import numpy as np
from scipy.stats import norm, lognorm
import seaborn as sns

# Fetch AAPL historical data
aapl_data = AAPL_data['Close']

# Calculate simple returns
simple_returns = AAPL_data['Simple Returns']

# Plot histogram of AAPL simple returns using seaborn
plt.figure(figsize=(10, 6))
sns.histplot(simple_returns, bins=30, kde=True, stat="density", color='skyblue',

# Fit normal distribution to simple returns
mu, std = norm.fit(simple_returns)
xmin, xmax = plt.xlim()
x = np.linspace(xmin, xmax, 100)
p = norm.pdf(x, mu, std)
plt.plot(x, p, 'k', linewidth=2, label='Normal Distribution Fit')
```

```
# Fit log-normal distribution to simple returns
# Note: lognorm expects the data to be in the original space, not simple returns
shape, loc, scale = lognorm.fit(np.exp(simple_returns), floc=0)
y = np.linspace(simple_returns.min(), simple_returns.max(), 100)
plt.plot(y, lognorm.pdf(np.exp(y), shape, loc, scale), 'r', linewidth=2, label='

plt.title('Histogram of AAPL Simple Returns with Best-Fit Distributions')
plt.xlabel('Simple Return')
plt.ylabel('Density')
plt.legend()
plt.grid(True)
plt.show()
```




```
import scipy.stats as stats
params = stats.norm.fit(AAPL_data['Simple Returns'])
d, p_value = stats.kstest(AAPL_data['Simple Returns'], 'norm', args=params)
print(f"KS Statistic: {d}, P-Value: {p_value}")

if p_value > 0.05:
    print('Simple Returns fits the Normal-distribution (fail to reject H0)')
```

```
else:
    print('Simple Returns does not fit the Normal-distribution (reject H0)')
```

```
KS Statistic: 0.07133605317682465, P-Value: 1.0152935369396603e-05
Simple Returns does not fit the Normal-distribution (reject H0)
```

```
params = stats.norm.fit(AAPL_data['Log Returns'])
d, p_value = stats.kstest(AAPL_data['Log Returns'], 'norm', args=params)
print(f"KS Statistic: {d}, P-Value: {p_value}")

if p_value > 0.05:
    print('Log Returns fits the Normal-distribution (fail to reject H0)')
else:
    print('Log Returns does not fit the Normal-distribution (reject H0)')
```

```
KS Statistic: 0.07392354321471126, P-Value: 4.129028509587428e-06
Log Returns does not fit the Normal-distribution (reject H0)
```

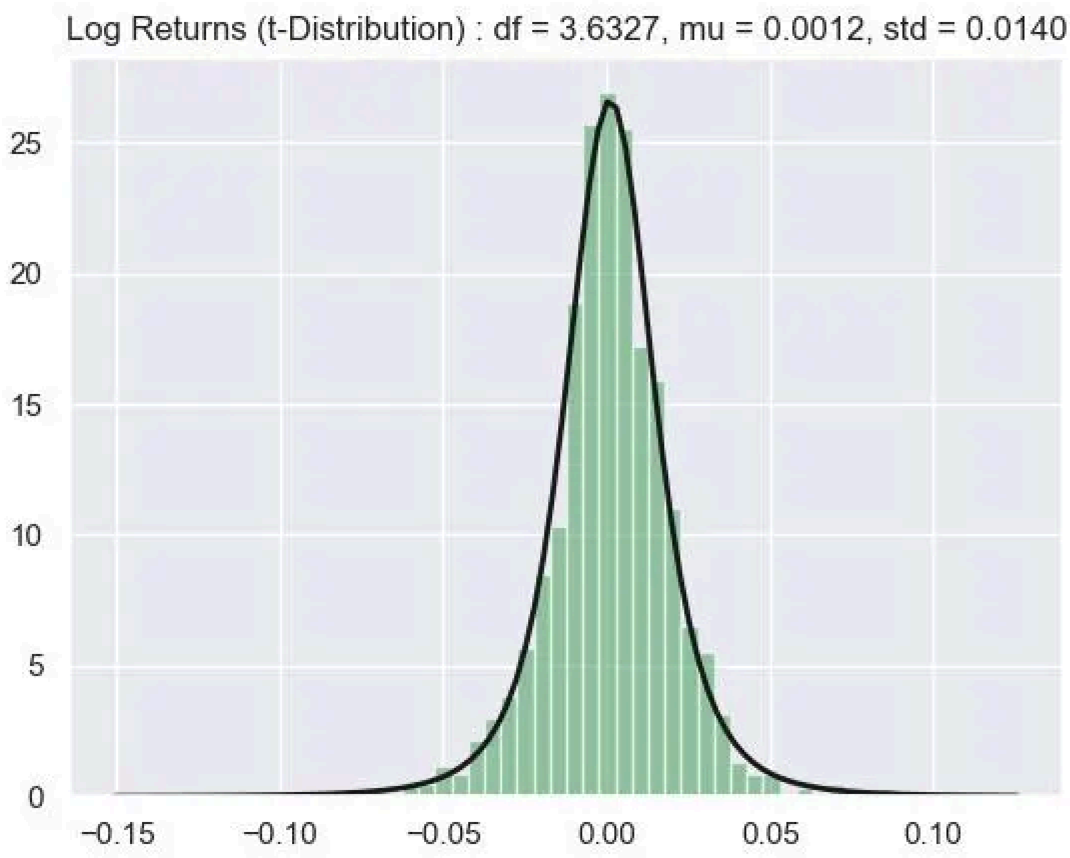
```
# Fit the data using t-distribution
params = stats.t.fit(AAPL_data['Simple Returns'])
d, p_value = stats.kstest(AAPL_data['Simple Returns'], 't', args=params)
print(f"KS Statistic: {d}, P-Value: {p_value}")

if p_value > 0.05:
    print('Sample Returns fits the t-distribution (fail to reject H0)')
else:
    print('Sample Returns does not fit the t-distribution (reject H0)')
```

```
KS Statistic: 0.017275132187029335, P-Value: 0.8627304723225454
Sample Returns fits the t-distribution (fail to reject H0)
```

```
# Fit the data using t-distribution
params = stats.t.fit(AAPL_data['Log Returns'])
d, p_value = stats.kstest(AAPL_data['Log Returns'], 't', args=params)
print(f"KS Statistic: {d}, P-Value: {p_value}")

if p_value > 0.05:
    print('Log Returns fits the t-distribution (fail to reject H0)')
else:
    print('Log Returns does not fit the t-distribution (reject H0)')
```

```
KS Statistic: 0.01634908528427123, P-Value: 0.9022649618254373
Log Returns fits the t-distribution (fail to reject H0)
```

```
# Fit a t-distribution to the log return data
params = stats.t.fit(AAPL_data['Log Returns'])

# Get the parameters
dfree, mu, std = params
print(f"Degrees of Freedom: {dfree}, Mean: {mu}, Standard Deviation: {std}")

# Plot the histogram and the PDF
plt.hist(AAPL_data['Log Returns'], bins=50, density=True, alpha=0.6, color='g')

xmin, xmax = plt.xlim()
x = np.linspace(xmin, xmax, 100)
p = stats.t.pdf(x, dfree, mu, std)

plt.plot(x, p, 'k', linewidth=2)
title = f"Log Returns (t-Distribution) : df = {dfree:.4f}, mu = {mu:.4f}, std =
plt.title(title)

plt.show()
```

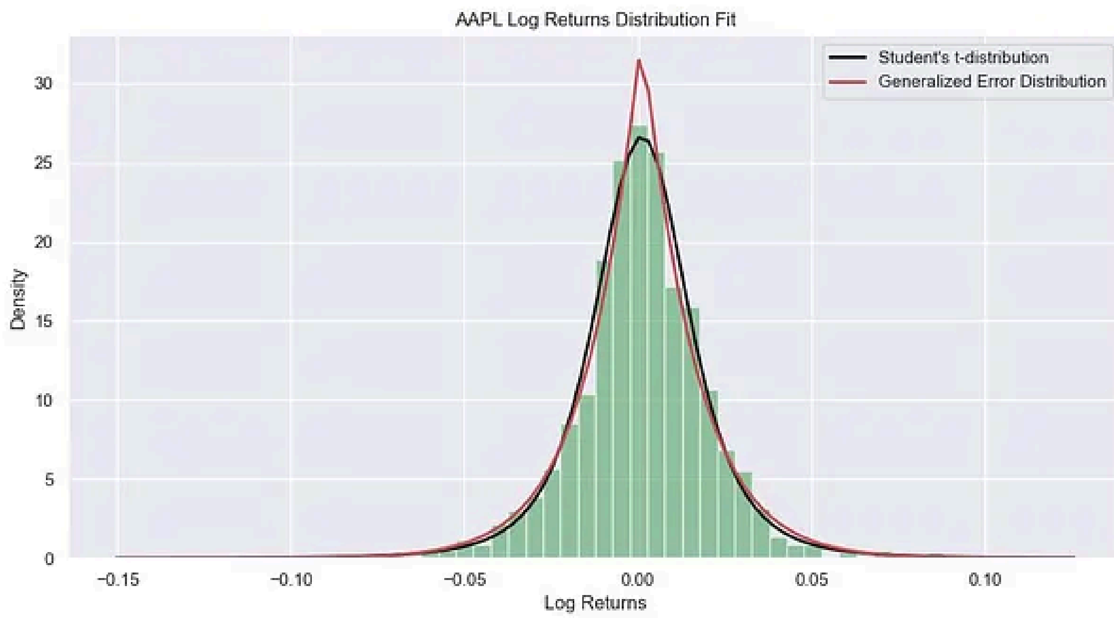


New update (2024–12–27):
After reading this article (Exploring Returns: Beyond Normal with Generalized Error Distribution), I wonder whether the log returns fit better with a Student's t-distribution or a Generalized Error Distribution

```
import yfinance as yf
import scipy.stats as stats
from scipy.stats import t, gennorm

endDate = "2024-10-01"
startDate = "2020-01-01"

ticker ='AAPL'
```

```
# Download AAPL stock data
aapl = yf.download(ticker, start=startDate, end=endDate)

# Calculate daily log returns
aapl['Log Return'] = np.log(aapl['Adj Close'] / aapl['Adj Close'].shift(1))

log_returns = aapl['Log Return'].dropna()

# Fit Student's t-distribution
t_params = stats.t.fit(log_returns)

# Fit Generalized Error Distribution
ged_params = stats.gennorm.fit(log_returns)

# KS test for Student's t-distribution
ks_stat_t, p_value_t = stats.kstest(log_returns, 't', args=t_params)

# KS test for Generalized Error Distribution
ks_stat_ged, p_value_ged = stats.kstest(log_returns, 'gennorm', args=ged_params)

print(f"KS Statistic for Student's t-distribution: {ks_stat_t}, p-value: {p_valu
print(f"KS Statistic for Generalized Error Distribution: {ks_stat_ged}, p-value:

# Plot histogram and PDFs
plt.figure(figsize=(12, 6))

# Histogram of log returns
plt.hist(log_returns, bins=50, density=True, alpha=0.6, color='g')

# PDF of Student's t-distribution
xmin, xmax = plt.xlim()
x = np.linspace(xmin, xmax, 100)
p_t = stats.t.pdf(x, *t_params)
plt.plot(x, p_t, 'k', linewidth=2, label='Student\'s t-distribution')

# PDF of Generalized Error Distribution
p_ged = stats.gennorm.pdf(x, *ged_params)
plt.plot(x, p_ged, 'r', linewidth=2, label='Generalized Error Distribution')

plt.title('AAPL Log Returns Distribution Fit')
plt.xlabel('Log Returns')
plt.ylabel('Density')
plt.legend()

plt.show()
```

```
KS Statistic for Student's t-distribution: 0.015197289912292966, p-value: 0.9417
KS Statistic for Generalized Error Distribution: 0.018040429842028582, p-value:
```



Interpretation

- KS Statistic: A lower KS statistic indicates a better fit.
- p-value: A higher p-value indicates that the data is more likely to come from the fitted distribution.

Thank you for reading.
Please let me know if anyone is familiar with stochastic processes related to Student's t-distribution (or Cauchy distributions or *Laplace distribution).*

Other related articles:

[Simple or Log Returns?]

[Returns and Log Returns]

[Understanding Financial Data: Simple vs. Log Returns & Normal Distribution]

[Analysis on Stocks: Log(1+return) or Simple Return?]

[Simple Returns vs. Log Returns: A Comprehensive Comparative Analysis for Financial Analysis]

[Why we use log returns for stock returns]

[Why We Use Log Return in Finance?]

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A fervent enthusiast, fueled by an insatiable curiosity for STEM (Science, Technology, Engineering, and Mathematics) disciplines.